DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

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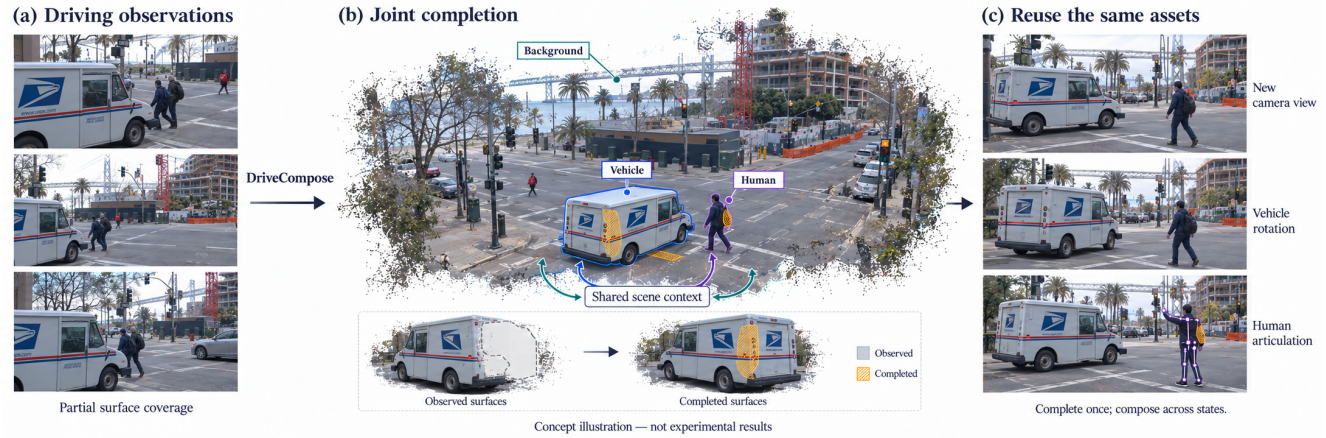


Figure 1. **DriveCompose concept.** DriveCompose jointly completes background, vehicle, and human content from driving images to form a compositional Gaussian scene. The same assets are reused under changes in camera viewpoint, vehicle pose, and human articulation. Gold marks completed regions.

Abstract

Reconstructing complete and controllable 3D scenes from partial observations is essential for autonomous driving simulation and scene editing. We present DriveCompose, a framework for generative reconstruction of compositional driving scenes from image sequences. We represent the background in world coordinates and vehicles and humans in their respective canonical frames, allowing completed surfaces to follow vehicle motion and human articulation. A conditional flow-matching model jointly completes missing background and actor content in these representations, using cross-component memory to share scene context. To maintain consistency with input observations, we preserve observed surface geometry and appearance during completion, then render the completed scene at recorded camera viewpoints and actor states for comparison with the input images. On Waymo, DriveCompose improves dynamic scene reconstruction and off-trajectory view synthesis over the evaluated baselines. Compared with the variant without cross-component memory, our model reduces mean absolute RGB error on unseen surfaces by 20.5% in component ablations. The resulting Gaussian scene supports composition and rendering with the same completed assets under changes in camera viewpoint, vehicle pose, and human articulation.

1. Introduction

Driving simulation and scene editing require 3D scenes that support changes in camera viewpoint, vehicle motion, and human pose [46, 49]. However, driving images reveal only the surfaces visible from the recorded viewpoints and actor configurations [1, 31]. Moving a camera, turning a vehicle, or changing a pedestrian's pose exposes previously unobserved regions. Reconstructing scenes for these operations therefore requires completing the missing geometry and appearance while ensuring that completed vehicle and human surfaces move with their respective actors.

Recent feed-forward methods fuse multiple images to directly predict driving-scene geometry and Gaussian representations [3, 6, 45, 48]. Motion estimation and visibility

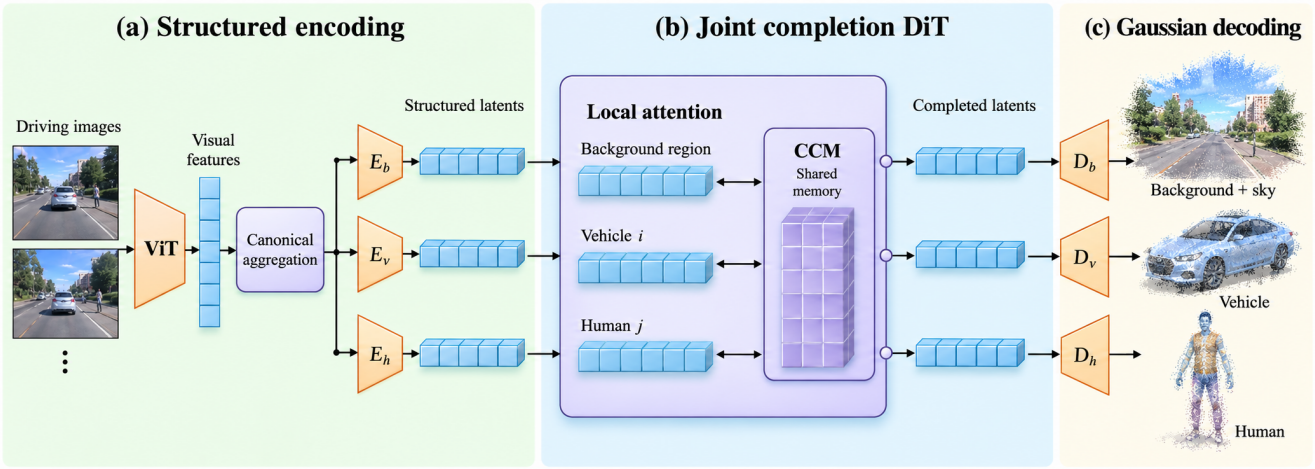


Figure 2. **DriveCompose overview.** (a) A shared ViT extracts visual evidence, which is aggregated on background and canonical asset supports and encoded into structured latents. (b) Joint completion applies local attention within groups and shares context through cross-component memory (CCM). (c) Type-specific decoders produce the Gaussian background, canonical vehicle and human assets, and a sky map.

modeling further help align dynamic objects and integrate observations over time. Multi-frame aggregation increases coverage of surfaces visible in the input sequence. When a new viewpoint or actor motion reveals surfaces unseen throughout the sequence, reconstruction must infer their geometry and appearance from generative priors.

Generative reconstruction uses asset priors, Gaussian completion, or generated views to infer unobserved content [17, 44, 47, 51]. Meanwhile, compositional reconstruction represents the background and moving actors separately to support vehicle motion and human pose control [5, 43, 49]. Applying generative completion to such scenes requires generated surfaces to move with their actors, while allowing each actor's completion to use information from the surrounding scene.

To this end, we introduce DriveCompose, which reconstructs driving image sequences into a generatively completed 3D scene with independently controllable actors. We represent the background in world coordinates, vehicle surfaces in each vehicle's own coordinate frame, and human surfaces in a reference pose with skeletal bindings [5, 43]. Completion operates directly on these representations, allowing generated vehicle and human surfaces to follow their actors' motion and pose changes (Sec. 3.1). The model jointly completes missing background and actor content using shared scene context.

As illustrated in Fig. 2, DriveCompose first recovers scene geometry and actor motion from the image sequence and aligns multi-frame image features with the background, vehicle, and human representations. The completion model captures local structure within each component, while cross-component memory gathers information from the background and actors and distributes the shared context back to each component. Missing regions of a vehicle or human can therefore be completed using information from the surrounding scene. The model learns to generate missing latent codes through flow matching [18], and component-specific decoders convert the completed representations into a Gaussian scene [15].

During completion, we retain surfaces established by reliable observations and generate the missing geometry and appearance. Newly generated surfaces may occlude previously visible regions, so we render the completed scene at the recorded camera viewpoints and actor states and compare it with the input images to check for inconsistencies introduced by completion. Scenes that pass this check can reuse the same completed content for rendering under different camera viewpoints and actor states (Sec. 3.5).

On Waymo, DriveCompose improves dynamic scene reconstruction and off-trajectory view synthesis over the evaluated baselines (Sec. 4). Component ablations further show that cross-component information sharing reduces unseen-surface completion error, with larger absolute error reductions under sparser observations (Sec. 4.4).

Our contributions are:

- We introduce DriveCompose, a generative reconstruction framework for closed-loop driving simulation. It recovers and completes backgrounds, vehicles, and humans from image sequences to construct compositional Gaussian scene assets. The resulting scene reuses the same completed assets for composition and rendering as camera poses, vehicle poses, and human poses change.
- We develop a cross-component latent completion model based on conditional flow matching. Local attention mod-

els spatial relationships within each component, while cross-component memory aggregates background and actor features to provide scene context for generating each component's missing content.
- We evaluate the framework and its key mechanisms through dynamic reconstruction, off-trajectory view synthesis, and component ablations. Experiments show that DriveCompose outperforms the evaluated baselines on reconstruction and view synthesis; cross-component information sharing reduces unseen-surface completion error, with larger absolute reductions under sparse observations.

Section 2 reviews related work, Section 3 presents the method, and Section 4 reports experiments. Sections 5 and 6 discuss limitations and conclude.

2. Related work

Driving-scene reconstruction. NeRF and Gaussian splatting provide the radiance representations underlying urban reconstruction [15, 22], with driving-specific rendering developed in NeuRAD and SplatAD [11, 33]. Feed-forward methods learn geometry and Gaussian prediction from images [2, 4, 12, 35, 36]. For driving logs, STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with temporal visibility and motion-guided interpolation [3, 45]. DrivingForward, StreetForward, PointForward, and Instant NuRec further explore surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction [6, 14, 32, 48]. TokenGS also decouples Gaussian prediction from pixels through learnable tokens [29]. DriveCompose aggregates recovered evidence onto background and canonical asset supports, then conditionally generates their missing content before Gaussian decoding.

Compositional scenes and controllable assets. Neural Scene Graphs, Street Gaussians, and DrivingGaussian separate scene content into background and moving components [23, 43, 50]. OmniRe includes canonical representations for rigid and articulated actors, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [5, 46, 49]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [13, 16, 20, 25, 27]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [21, 28, 52]. These works establish the representation and control mechanisms used to manipulate scene components. Our completion variables retain this component structure, so generated surfaces inherit an asset identity, coordinate frame, and motion binding.

Generative reconstruction and completion. Image-conditioned 3D generation uses generated views [9, 19, 30] or structured latents encoding geometry and appearance [41, 42]. GenAssets learns driving-asset priors, CompleteSplat completes hidden Gaussian content, and JRM aggregates unaligned object observations through latent flow matching [17, 40, 47]. LatentWorld organizes semantic occupancy generation around entity and background latents [24], making structured scene generation an important point of comparison. DriveCompose targets RGB-renderable reconstruction conditioned on driving logs: a shared completion model exchanges context across world-space background and canonical vehicle and human codes. Observed codes and known free space constrain sampling, and original-view rendering checks the assembled result before reuse.

Generative driving scenes and videos. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [8, 10, 37]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction; HorizonForge combines scene editing with video diffusion [38, 44]. WorldSplat generates Gaussians and refines their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [39, 51]. These methods connect generative imagery and three-dimensional scene representations at different stages. DriveCompose performs completion during scene construction and stores the resulting background and canonical assets; subsequent camera, rigid-body, and skeletal states render the same completed content.

3. Method

DriveCompose reconstructs a driving scene in the units that will later be controlled. Figure 2 shows the main reconstruction pipeline.

3.1. Problem formulation and compositional representation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. Rig poses are provided or estimated relative to the first frame. Coordinate conventions and image preprocessing are detailed in Appendix D of the supplementary material.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian

splatting [15]. Vehicles retain local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The decoder also carries forward the fixed support, identity, binding, and sky-validity metadata in Eq. 1. The layout is established from the input log; generation fills its unknown content. Simulation time changes the state of the decoded scene, while flow time indexes the construction of its latent codes.

3.2. Visual recovery and canonical aggregation

Shared visual encoding. The Driving Log Encoder uses transformer features [7, 34] to predict camera-axis depth, normals, confidence, semantics, and instance assignments from the calibrated image log. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Architectural details and state parameterization are given in Appendix D of the supplementary material.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; vehicle anchors are motion-normalized and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The query confidence is the maximum over accepted observations. A reliable ray votes for free space only when the entire support cell is in front of the measured surface; the criterion is detailed in Appendix D of the supplementary material.

Positive accumulated observation weight marks a query as observed. Otherwise, sufficient free-space votes mark it free, and the remainder is unknown. The masks m_i, f_i, and u_i are mutually exclusive and satisfy $m_i + f_i + u_i = 1$. Sky projection uses camera rotation alone, reliable sky semantics, and zero depth residual; sky queries receive no free-space votes.

3.3. Structured latent representation and decoding

Type-specific encoders map $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ to codes $\mathbf{z}_i \in \mathbb{R}^D$. Each query keeps its type, ownership, and coordinate frame even though all codes have the same dimension. Visibility masks and confidence are supplied separately to the completion model. Partial and complete observations therefore use the same intrinsic encoding map. Before conditioning, unobserved codes are set to zero; the observed subset is denoted $\mathbf{z}^{\text{obs}}$.

Background, vehicle, and human decoders independently map each query to K child Gaussians confined to its support cell. Human children inherit the query's skinning weights; vehicle children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}. Gaussian and appearance parameterizations are detailed in Appendix E of the supplementary material.

Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

3.4. Conditional generative reconstruction

Generative completion module. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling. We parameterize its completion velocity with a DiT-style transformer [26] trained by flow matching [18]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. A cross-component memory (CCM) module then exchanges context through shared memory:

$$\mathbf{M}' = \mathbf{M} + \mathrm{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \mathrm{Attn}_w(\mathrm{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates locally updated tokens, and attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters the local normalization; learned tokens and log summaries initialize memory. This structure keeps each support's local organization while allowing a vehicle or human query to use information from the surrounding scene.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\mathrm{obs}} + u \odot [(1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\mathrm{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

All unknown queries interpolated in Eq. 8 require valid target codes, even if s_i excludes some of them from the loss. This requirement determines how complete targets are constructed in Sec. 3.6.

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{10}$$

The projection $\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed fixes one sampled completion. Sampling precedes decoding, and the resulting scene is reused when the camera or actor states change.

3.5. Scene composition and observation consistency

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, vehicle transforms, and human root and joint poses. Vehicles are placed by rigid transforms and humans by skinning; standard transformations and backend details are given in Appendix C of the supplementary material. The renderer composites the background, active assets, and valid sky:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \tag{11}$$

Resetting or branching a trajectory changes state while preserving $\mathcal{W}$ and its seed. Finite support and supported sky directions determine coverage.

Checking a constructed scene. Fixed observed codes preserve decoded parameters, but a newly generated surface can still occlude them. Before accepting a scene, we render the completed candidate and an observed-only reference at the original cameras and actor states. The reference disables all non-observed queries. Let $e_{\mathrm{cand}}(p)$ and $e_{\mathrm{obs}}(p)$ be their mean absolute RGB errors at pixel p. We measure local degradation with

$$\Delta e(p) = \max\{e_{\mathrm{cand}}(p) - e_{\mathrm{obs}}(p), 0\}. \tag{12}$$

Averaging this quantity over reliable non-ego pixels prevents improvements in one region from cancelling degradation elsewhere. Acceptance also bounds absolute RGB and valid-depth errors, minimum coverage, and foreground intrusion, using aggregate and worst-view checks. An intrusion is a newly rendered surface in front of the predicted reference depth that is absent from the observed-only render; its rate is normalized over reliable non-sky geometry pixels. Predicted depth provides an observation-consistency reference. Independent geometry evaluation requires a separate reference. A rejected candidate is not exported for reuse.

3.6. Training procedure

Learning is divided into four stages so that the completion target has a stable meaning (Table 1).

For codec fitting, partial input and more complete observations share the same query positions, types, order, and ownership. With recovery and the entire codec frozen, complete scene or asset examples provide a valid target for every unknown query to train the GCM with Eq. 9. Real-log adaptation differentiates through flow integration, decoding, and rendering to update only the GCM. Complete-target samples

Table 1. Training schedule. R: recovery; C: codec; G: GCM. T: trained; F: frozen; –: unused.

Stage	R	C	G	Supervision
Recovery	T	–	–	Geometry, semantics, identity, states
Codec	F	T	–	Partial / complete observations
Prior	F	F	T	Complete targets
Adaptation	F	F	T	Real-log rendering and scene constraints

can be interleaved to retain the prior, with each loss applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration. Detailed supervision, losses, and conditioning perturbations are provided in Appendix B of the supplementary material.

At inference, a log is encoded once; its support and evidence remain fixed during completion, and the decoded scene is checked before reuse under requested states (Section 3.5).

4. Experiments

We evaluate our method along three axes: *(i)* dynamic scene reconstruction quality on Waymo Open Dataset (§ 4.2); *(ii)* off-trajectory view synthesis under lateral camera shifts (§ 4.3); and *(iii)* ablations of the main design choices (§ 4.4). Additional cross-dataset generalization results are provided in the supplementary material.

4.1. Experimental Setup

Datasets and protocol. We use the Waymo Open Dataset with 798 training and 202 validation sequences. Following STORM, each validation sequence is divided into non-overlapping 2-second clips of 20 frames at 10 Hz [45]. The FRONT, FRONT_LEFT, and FRONT_RIGHT cameras provide observations at zero-based frame indices $\{0, 5, 10, 15\}$. The remaining 16 frames are used for evaluation: 12 frames for temporal interpolation and four frames for prediction beyond the last observation. Predictions and reference images are resized to 240×160 pixels for evaluation [14].

Baselines. We compare with STORM, DGGT, StreetForward [3, 45, 48]. Methods use the same context timestamps and target views, with model-specific preprocessing. Camera parameters, additional geometric inputs, and rendering adapters are documented with the implementation settings. For spatial extrapolation, we compare Gaussian renderings before diffusion refinement, following PointForward [6].

Metrics. We report PSNR and SSIM over the full image and shared annotated regions. Depth quality is measured using AbsRel and δ_1 at valid projected LiDAR samples after per-frame scale alignment [14]. Scores are averaged over frames within each clip and then over clips within each sequence; sequences receive equal weight. Novel-trajectory renderings are evaluated using FID and FVD against a fixed real-image and video reference population.

Implementation details. DriveCompose reconstructs category-specific actor assets from motion-aligned observations and composes them with a static background. Reconstruction time includes preprocessing, asset generation, and scene composition. Rendering time is measured separately, and asset reuse is evaluated across repeated scene queries.

4.2. Dynamic Scene Reconstruction

We evaluate reconstruction at unobserved times along the recorded camera trajectory. Table 2 reports full-image, dynamic-region, and depth quality; Fig. 3 compares rendered appearance across five examples.

Quantitative comparison. DriveCompose achieves the best scores across all six metrics in Table 2. Full-image and dynamic-region PSNR exceed the strongest baseline in each category by 1.154 and 2.340 dB, respectively. The corresponding SSIM scores reach 0.901 and 0.862, while depth AbsRel decreases to 0.095 and δ_1 reaches 0.877.

Qualitative comparison. Figure 3 shows that DriveCompose retains the pedestrian silhouette in the first row and the rear profile of the central vehicle in the second row, while preserving the surrounding street layout. The baseline renderings exhibit displaced or repeated actor contours in the upper rows and blurred or distorted road structure in the lower rows. DriveCompose more closely follows these reference structures in the displayed examples, with residual artifacts around the pedestrians in the third row and the parked pickup in the last row.

Table 2. Dynamic scene reconstruction on Waymo. Image quality is reported over the full image and annotated regions; depth is evaluated at valid LiDAR samples.

	Full image		Dynamic Region		Depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.380	0.826	22.100	0.624	0.178	0.741
DGGT	27.410	0.794	22.710	0.651	0.249	0.668
StreetForward	27.010	0.818	20.919	0.592	0.197	0.759
Ours	**28.564**	**0.901**	**25.050**	**0.862**	**0.095**	**0.877**



Figure 3. **Qualitative comparison for dynamic scene reconstruction.** Each row shows one example. Columns, from left to right, are ground truth, DGGT, STORM, StreetForward, and DriveCompose (Ours). The examples cover pedestrians, traffic, and urban backgrounds, allowing comparison of actor silhouettes, vehicle boundaries, and road and building structure.

4.3. Off-Trajectory View Synthesis

We evaluate off-trajectory rendering under lateral shifts of ± 1, ± 2, and ± 4 meters, keeping camera orientation, intrinsics, and timestamps fixed [6, 51].

Table 3 reports FID and FVD averaged over separately evaluated left and right shifts at each magnitude, using the same real reference population across methods. Figure 4 presents qualitative view-extrapolation results on Waymo with the camera shifted 2 m to the left, illustrating vehicle shape, road layout, and occlusion boundaries across three scenes.

Table 3. Novel-view synthesis under lateral camera shifts. FID and FVD are averaged over left and right shifts at each displacement magnitude. Lower is better.

Method	Shift ± 1 m		Shift ± 2 m		Shift ± 4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

4.4. Ablation studies

Protocol. Paired evaluation of hidden surfaces is defined on independent complete-reference scenes; Waymo logs provide recorded-view evaluation. Each 29-timestamp window uses three synchronized cameras: front, front-left, and front-right. The nested input sets are $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$, with shared target timestamps $\{2, 6, \ldots, 26\}$. Target images are excluded from evidence and query construction; target camera and actor states are supplied equally to every variant. The recovery module and codec are frozen, with matched query layouts, data exposure, and optimization budgets at each input setting. Training samples $T \in \{2, 4, 8\}$ uniformly;



Figure 4. **Qualitative view extrapolation on Waymo with a 2 m leftward camera shift.** Each row shows one scene. Columns, from left to right, show the reference image (GT), STORM, DGGT, StreetForward, and DriveCompose (Ours).

completion uses 24 Euler steps. The two learned variants use training seeds $\{0, 1, 2\}$ and world seeds $\{7, 17, 27\}$; the variant w/o GCM is deterministic. Hyperparameters and acceptance thresholds are selected on an internal development split before evaluation on scene- and identity-disjoint data.

Metrics. New-MAE measures linear-RGB error on independently identified surfaces unseen in every input; NVS-MAE covers all held-out target pixels. Both MAE metrics and Obs-Reg use linear RGB on a $[0, 1]$ scale. Across input settings, New-MAE uses the common region unseen even under $\mathcal{T}_8$. Obs-Reg averages the positive input-view regression in Eq. 12. Cov. is the non-sky target fraction with support coverage > 0.5, opacity > 0.1, and finite positive depth. Uncovered pixels receive the same zero-RGB fill for error measurement. Pass is the fraction of all requested worlds accepted by the fixed checker, including construction failures in its denominator. Image metrics include every renderable candidate, including rejected worlds; non-renderable counts are recorded separately. Metrics are averaged within each scene and then across scenes and training seeds.

Completion and scene context. Table 4 compares the full model with two controls. The variant w/o GCM bypasses the generative completion module and disables the opacity of every non-observed, non-sky query. The variant w/o CCM replaces the cross-component memory module with isolated per-component memories at every layer, while sharing parameters and retaining the same global log context. The learned control is retrained through Stages 3–4; memory-related compute differences are recorded.

Table 4. **Component ablations.** All variants use $T = 8$ (24 images), shared recovery and codec, and common evaluation masks. Quality is measured before acceptance filtering; Pass includes all requested worlds. A dash denotes an unreported metric; bold marks the best reported value in each column.

Variant	New-MAE ↓	Obs-Reg ↓	Cov. (%) ↑	Pass (%) ↑
w/o GCM	0.186	–	74.6	**95.0**
w/o CCM	0.078	0.005	94.5	87.0
Ours	**0.062**	**0.003**	**96.3**	93.0

Removing completion raises New-MAE from 0.062 to 0.186 and reduces coverage from 96.3% to 74.6%. The variant w/o GCM has a 95.0% pass rate, but leaves substantially more target pixels uncovered. CCM reduces New-MAE from 0.078 (w/o CCM) to 0.062 (Ours), a 20.5% relative reduction, while increasing coverage by 1.8 percentage points and the pass rate by 6.0 points.

Input sparsity. Table 5 reuses checkpoints of the full model and the variant w/o CCM at $T = 2, 4, 8$, with 6/12/24 input images. The time span, target views, and common unseen-region mask remain fixed; evidence and queries are rebuilt from each input subset. No model is retrained for this comparison. For the full model, increasing T from 2 to 8 reduces NVS-MAE from 0.043 to 0.026 and New-MAE from 0.092 to 0.062, while increasing coverage from

Table 5. **Input-timestamp sensitivity.** At each T, we compare w/o CCM with the full model using fixed checkpoints and target views. New-MAE uses the common unseen region. Results at $T = 8$ match Table 4; bold marks the better value within each group.

T	Variant	NVS-MAE ↓	New-MAE ↓	Cov. (%) ↑
2	w/o CCM	0.052	0.119	88.5
	Ours	**0.043**	**0.092**	**91.7**
4	w/o CCM	0.038	0.094	92.3
	Ours	**0.032**	**0.074**	**94.6**
8	w/o CCM	0.030	0.078	94.5
	Ours	**0.026**	**0.062**	**96.3**

91.7% to 96.3%. The full model outperforms the variant w/o CCM on all three metrics at every input count. The absolute New-MAE reduction from CCM is 0.027, 0.020, and 0.016 at $T = 2, 4, 8$, respectively, while the corresponding coverage gains are 3.2, 2.3, and 1.8 percentage points. The larger absolute gains at lower input counts indicate that cross-component context is especially useful when observations are sparse.

4.5. State-driven reuse

At timestamp 14 and world seed 7, the interventions independently translate the camera by 1 m along its local $+x$ axis, rotate a vehicle by 30° about its root and the world vertical axis, or apply a held-out valid human pose H_1 with fixed root placement. Lighting and canonical assets remain fixed, without resampling completion.

5. Limitations

DriveCompose represents a finite region under fixed illumination. Errors in recovered depth, identity, or canonical pose can enter the observed anchors and constrain later completion incorrectly. Vehicles are rigid, and humans retain fixed clothing volume and skeleton binding; poses outside the template's support can expose missing geometry. Reference-world coordinates provide context at construction, while contact and collision validity after an edit depend on the physical backend. The observation check assesses agreement with recorded evidence. Geometry outside that evidence and appearance under changed illumination require separate supervision or modeling.

6. Conclusion

We presented DriveCompose, a generative reconstruction framework for compositional driving scenes. Its structured latent representation gathers image evidence into background and canonical assets, giving missing content a defined location, identity, and motion model before generation. Conditional completion uses scene context while retaining observed evidence, and typed decoding produces the components of a reusable Gaussian scene. Separating content construction from state-driven composition allows subsequent camera and actor changes to reuse the same completion.

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